

Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Mock Paper Set 1

Paper Reference **8MA0/21**

Mathematics Advanced Subsidiary Paper 21: Statistics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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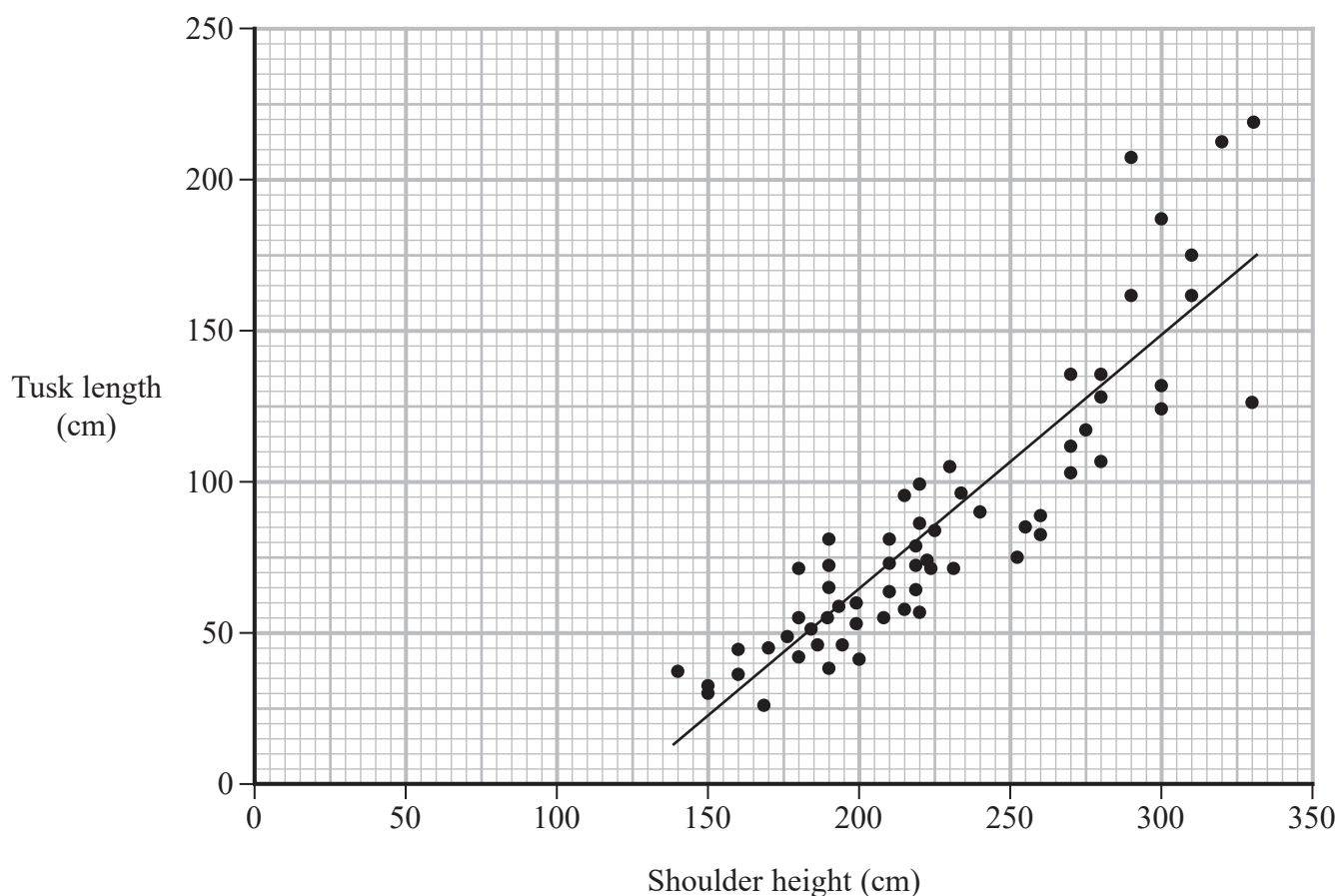


Pearson

1. Scientists collected data for some African elephants.

Matthew is studying the relationship between elephant shoulder height, s cm, and tusk length, t cm.

He uses only the data for male elephants and draws the following scatter diagram using statistical software.



- (a) Give an interpretation of the correlation between tusk length and shoulder height. (1)

The equation of the regression line of t on s for these data is $t = 0.843s - 104$

- (b) Give an interpretation of the gradient of this regression line. (1)

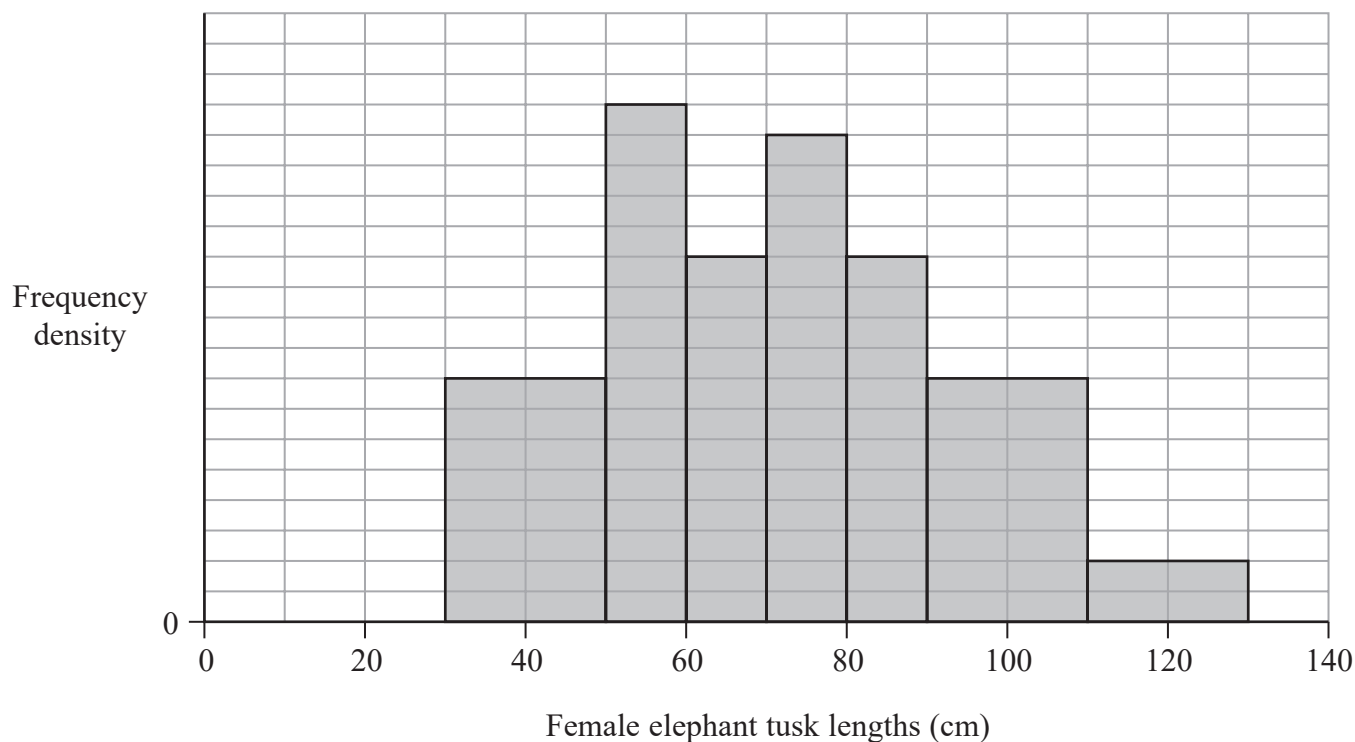
Matthew suggests using this regression line to draw conclusions about the relationship between elephant shoulder height and tusk length for **all** elephants.

- (c) Explain why his suggestion may not be appropriate. (1)



Question 1 continued

Matthew also draws the following histogram for the tusk lengths of female elephants.



(d) Estimate the percentage of female elephants with a tusk length of 75 cm to 100 cm.

(3)



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



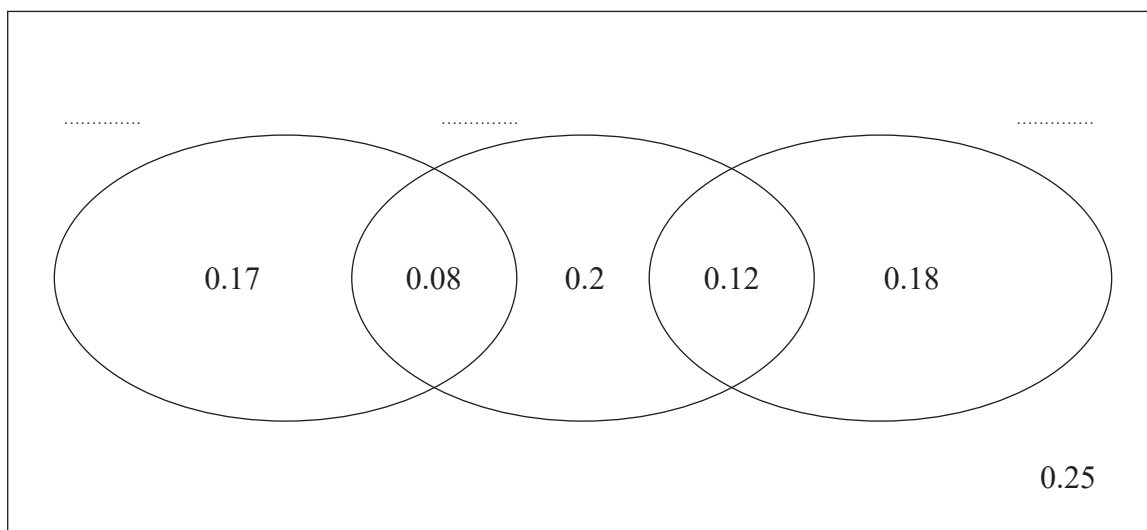
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2. The Venn diagram shows three events and the probabilities associated with each region.

The labels for the three events are missing.



The three events shown by the Venn diagram are A , B and C .

- Event A and event B are mutually exclusive
- Event A and event C are independent

- (a) Giving reasons and showing your working, complete the Venn diagram with the correct labels for A , B and C .

(4)

- (b) Write down the probability of event B not occurring.

(1)

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. Kathleen is investigating the weather for Heathrow from May to October 2015.

She uses the large data set and produces the summary statistics below for the daily mean temperature ($t^{\circ}\text{C}$).

$$n = 184 \quad \sum t = 2876.7 \quad \sum t^2 = 46\,797.3 \quad \text{Maximum} = 28.7$$

Kathleen defines limits for outliers for each variable using the mean ± 3 times standard deviation rule.

- (a) Show that $t = 28.7$ is an outlier.

(3)

Kathleen plans to produce summary statistics for the other weather variables recorded for Heathrow.

- (b) Giving a reason for your answer, state a numerical weather variable from the large data set for which her outlier rule will not give practically useful results.

(2)

Kathleen summarised the daily mean visibility for Heathrow from May to October 2015 in the table below.

Daily mean visibility (x decametres)	Number of days
$400 \leq x < 1200$	7
$1200 \leq x < 2000$	24
$2000 \leq x < 2400$	32
$2400 \leq x < 2800$	40
$2800 \leq x < 3200$	30
$3200 \leq x < 3600$	26
$3600 \leq x < 4400$	25

- (c) Use linear interpolation to find an estimate of the lower quartile.

(2)



Question 3 continued

Another student, Mason, calculated the mean and standard deviation of the daily mean temperature ($^{\circ}\text{F}$) in Jacksonville from May to October 2015.

$$\text{Mean} = 76.7^{\circ}\text{F}$$

$$\text{Standard deviation} = 4.86^{\circ}\text{F}$$

The formula to convert C degrees Celsius to F degrees Fahrenheit is $F = 1.8C + 32$

(d) Find, in degrees Celsius,

(i) the mean

(ii) the standard deviation

of the daily mean temperature in Jacksonville from May to October 2015.

(2)



Question 3 continued

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Question 3 continued

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(Total for Question 3 is 9 marks)



4. A company produces blue beads and green beads.

The proportion of green beads produced is 0.38

Beads are selected at random and put into mixed packs of 10

- (a) Find the probability of a pack containing more than 6 green beads. (2)
- (b) Find the most likely number of green beads in a pack.
Show your working clearly. (2)

The company also produces bowls.

Initial tests suggest that 20% of these bowls are faulty.

The manager of the company believes that the proportion of faulty bowls is greater than 20%

He takes a random sample of 40 bowls.

- (c) Using a 5% level of significance, find the critical region for a one-tailed test to check the manager's belief.
State your hypotheses clearly. (3)
- (d) Find the actual level of significance for this test. (1)

The manager finds that 17 of the 40 bowls sampled are faulty.

- (e) With reference to this information and part (c), comment on the manager's belief. (1)
- (f) Comment on the validity of the statistical model used in part (c), giving a reason for your answer. (1)



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Question 4 continued

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Question 4 continued

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 10 marks)

TOTAL FOR STATISTICS IS 30 MARKS

